Domain name system :
DNS Name Space, Resource records, Name Servers.

Domain name system

The Domain Name System (DNS) is an application-layer protocol that translates human-readable domain names (e.g., `www.google.com`) into IP addresses (e.g., `142.250.183.14`). It is a distributed and hierarchical naming system.

DNS Name Space:

DNS name space is organized in a hierarchical tree structure. The DNS Name Space is a hierarchical, distributed naming system used to organize and identify computers and other resources on the Internet. It maps domain names to IP addresses.

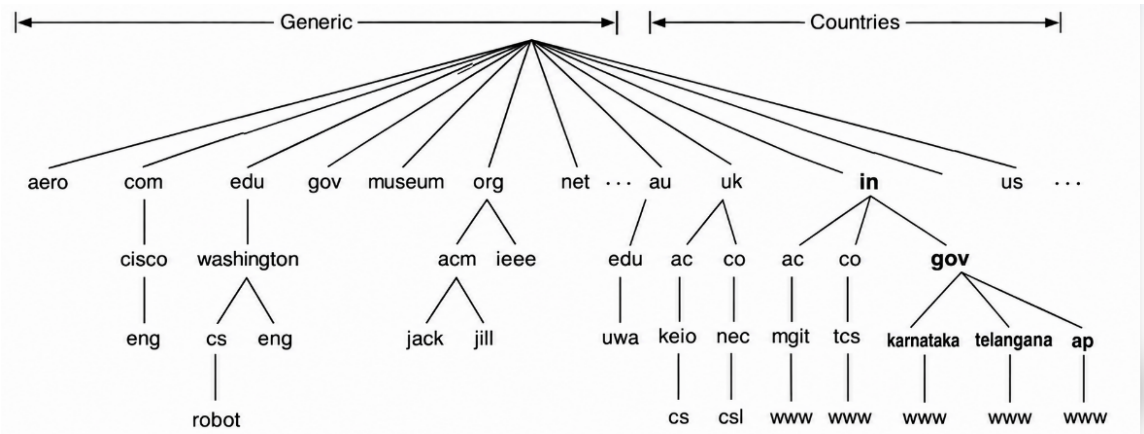


Figure 7-1. A portion of the Internet domain name space.

It consists of the following levels:

1. **Root Domain (.)**

- The highest level in the DNS hierarchy.
- Represented by a dot (.).
- It is managed by root name servers.

2. **Top-Level Domain (TLD):** the Internet is divided into over 250 top-level domains, where each domain covers many hosts. Each domain is partitioned into subdomains, and these are further partitioned, and so on. The top-level domains are run by registrars appointed by ICANN (Internet Corporation for Assigned Names and Numbers).Located directly below the root.

- Two main types:
    - **Generic TLDs:** Generic domains are used to classify websites and organizations based on their purpose or type. They help users identify the nature of an organization from its domain name. They are named using short abbreviations that indicate the organization's category or purpose. Example `mgit.ac.in` is the domain name of an academic institution in India.

Domain	Intended use
com	Commercial
edu	Educational institutions
gov	Government
int	International organizations
mil	Military
net	Network providers
org	Non-profit organizations
aero	Air transport

- Country-code TLDs: The country domains include one entry for every country, as defined in ISO 3166. Internationalized country domain names that use non-Latin alphabets were introduced in 2010. Example `mgit.ac.in` is the domain name of an academic institution in India
  - Example `.in`, `.uk`, `.us`

3. **Second-Level Domain :** A second-level domain (e.g., `company.com`) can be obtained easily by contacting a registrar authorized by ICANN. The registrar checks whether the requested domain name is available and does not violate any trademark. If there are no issues, the applicant pays a small annual fee and registers the domain name. It is Located below the top-level domain.Usually represents an organization or company name.

- Examples: `google.com`, `mgit.ac.in`

4. Subdomain: A subdomain is a subdivision of a main domain used to organize different departments, services, or sections of an organization.It is created by adding a prefix to the main domain name.

Example: `cse.mgit.ac.in`

5**. Host Name:** A host name is a name assigned to a specific computer or server connected to a network. It identifies a particular device or service within a domain.

Example: www : Host name (Web server)

- `mail.google.com` – Gmail webmail service
- `smtp.gmail.com` – SMTP server for sending emails
- `imap.gmail.com` – IMAP server for receiving and accessing emails
- `pop.gmail.com` – POP server for downloading emails

Resource records : Resource Records (RRs) are entries stored in DNS databases that provide information about domain names, such as IP addresses, mail servers, and name servers.

Each resource record contains the following fields:

Field	Description
Name	Domain name to which the record belongs
Value	Information associated with the domain name
Type	Specifies the type of resource record

Class	Specifies the network type (usually IN for Internet)
TTL	Time To Live; indicates how long the record can be cached
Length	Specifies the length of the resource data

Types of Resource Records

Type	Description	Example
A	Maps domain name to IPv4 address	`google.com → 142.250.183.14`
AAAA	Maps domain name to IPv6 address	IPv6 address
NS	Specifies the authoritative name server	`ns1.google.com`
MX	Specifies the mail server	`mail.google.com`
CNAME	Specifies an alias for another domain	`www.example.com`
PTR	Maps IP address to domain name (reverse lookup)	IP → hostname
SOA	Contains administrative information about a DNS zone	Primary DNS server

Example of a Resource Record:

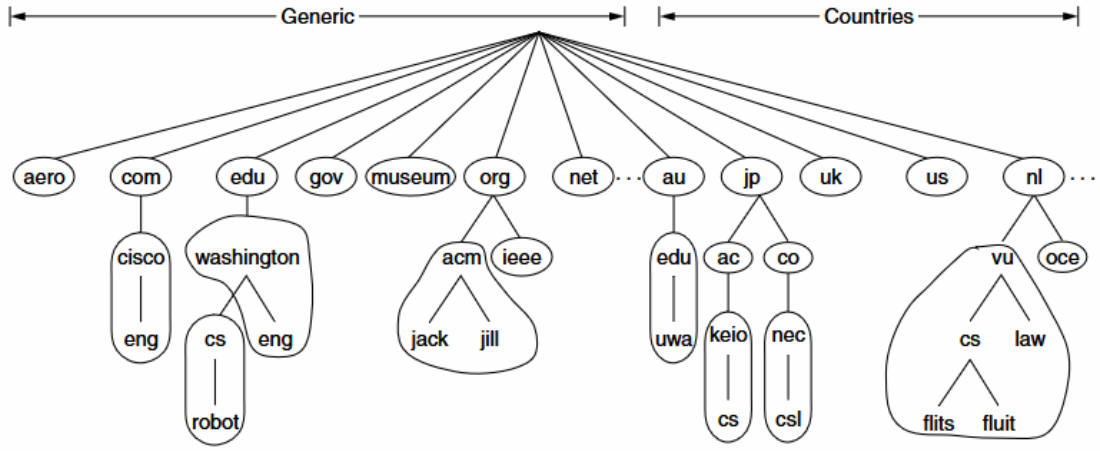
```
google.com IN A 142.250.183.14
```

- `google.com` – Name
- `IN` – Class (Internet)
- `A` – Type (IPv4 address)
- `142.250.183.14` – Value (IP address)

Name Server:

A name server is a computer that stores DNS records and provides information about domain names, including their corresponding IP addresses. Although a single name server could store the entire DNS database and respond to all DNS queries, it would become overloaded due to the enormous number of requests.

To overcome this problem, the DNS namespace is divided into **non-overlapping zones**. Each zone contains a specific portion of the DNS hierarchy and is managed independently by its own administrator and name servers. This division distributes the workload, improves reliability, and simplifies DNS management. A possible division of the DNS namespace into zones is shown in the figure, where each circled region represents a separate zone. Each zone has one or more DNS name servers that store its DNS records.



A zone is a portion of the DNS namespace managed by a particular administrator. `.edu` branch in the diagram: The University of Washington owns the domain `washington.edu`. However, it can divide its domain into smaller zones. The English Department does not want to maintain its own DNS server. Therefore, `eng.washington.edu` is managed by the University of Washington's DNS administrators. It belongs to the washington.edu zone. The Computer Science Department wants to maintain its own DNS server. Therefore, the university delegates `cs.washington.edu` to the CS department. It becomes a separate zone, managed by its own administrator and name servers.

```
washington.edu
      |
     eng
```

```
washington.edu
      |
      cs
      |
    robot
```